

Environment setup

1. Install SDK Driver

In the provided source package, extract **YDLidar-SDK-master.zip** to get the **YDLidar-SDK** folder. This folder contains the SDK files for the LiDAR. Since the ROS package for this LiDAR requires the SDK to be installed in advance, the driver files are stored inside the YDLidar-SDK folder. Open a terminal in that folder and run:

```
mkdir build
cd build
cmake ..
make -j4
sudo make install
```

If no errors appear during the process, the driver has been installed successfully.

2. Create Workspace & Build Packages

- **Option 1:** Extract the source package `yahboomcar_ws` to your home directory, then build it directly using `colcon build`:

```
cd yahboomcar_ws
colcon build
```

After a successful build, add the workspace path to `.bashrc`:

```
sudo gedit ~/.bashrc
```

Append the following line to the end of the file:

```
source ~/yahboomcar_ws/install/setup.bash --extend
```

- **Option 2:** Create a custom-named workspace. Using `oradar_ws` as an example, run in the terminal:

```
mkdir oradar_ws
cd oradar_ws
mkdir src
cd src
catkin_init_workspace
```

Then copy the package folders from `yahboomcar_ws/src` into `oradar_ws/src`. Navigate to the `oradar_ws` directory and build:

```
cd oradar_ws
colcon build
```

After a successful build, add the workspace path to `.bashrc`:

```
sudo gedit ~/.bashrc
```

Append the following line to the end of the file:

```
source ~/oradar_ws/devel/setup.bash --extend
```

3. Bind LiDAR Port Name

Open a terminal in the `yahboomcar_ws` workspace and run the following commands:

```
sudo chmod 777 src/ydlidar_ros2_driver-humble/startup/*
sudo sh src/ydlidar_ros2_driver-humble/startup/initenv.sh
```

```
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$ sudo chmod 777 src/ydlidar_ros2_driver-humble/startup/*
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$ sudo sh src/ydlidar_ros2_driver-humble/startup/initenv.sh
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$
```

Then unplug and re-plug the LiDAR serial port. Verify the port binding by running:

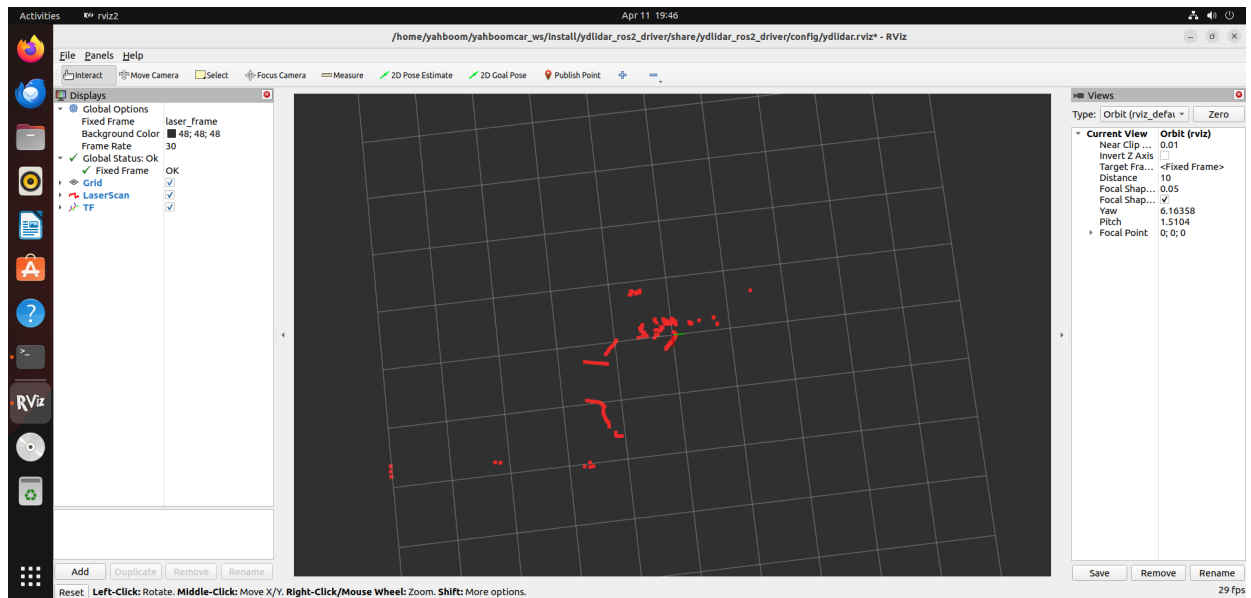
```
ll /dev/ydlidar
```

```
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$ ll /dev/ydlidar
lrwxrwxrwx 1 root root 7 Apr 11 19:44 /dev/ydlidar -> ttyUSB0
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$
```

4. Launch the LiDAR

Open a new terminal and enter the following command to start the LiDAR and display its output in RViz (**an external monitor is required**):

```
ros2 launch ydlidar_ros2_driver ydlidar_launch_view.py
```



To view the LiDAR node data via topic, run:

```
ros2 topic echo /scan
```

```
yahboom@yahboom-virtual-machine:~/yahboomcar_ws$ ros2 topic echo /scan
header:
  stamp:
    sec: 1712836021
    nanosec: 44564000
  frame_id: laser_frame
angle_min: -3.1415927410125732
angle_max: 3.1415927410125732
angle_increment: 0.015591030940413475
time_increment: 0.0002511637285351753
scan_time: 0.09971199929714203
range_min: 0.029999999329447746
range_max: 12.0
ranges:
- 0.0
- 0.2109999995470047
- 0.20299999941587448
- 0.202000000703334808
- 0.19799999989271164
- 0.195999999487400055
- 0.194999999284744263
- 0.0
- 0.19799999989271164
- 0.194999999284744263
```